

JACKSON BOPP

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Computer Engineering student seeking an entry-level test/QA or hardware validation engineering role, with hands-on experience in test automation, root cause analysis, and reliability engineering.

SKILLS

Testing & QA: GoogleTest/gmock, pytest, regression testing, root cause analysis, flaky test diagnosis, hardware-in-the-loop test automation, capture-and-replay tooling, schema validation

CI/CD & Tooling: GitHub Actions, Docker, Gerrit code review, clang-format, checkpatch, meson, west, git bisect

Languages: Python, C, C++, TypeScript, Bash, MATLAB, Verilog/VHDL

Systems: Linux, D-Bus, MQTT, PostgreSQL, Docker Compose, Raspberry Pi, MSP430, ESP32, STM32

Certifications: Certified SOLIDWORKS Associate (CSWA), CSWA Additive Manufacturing

EDUCATION

University of South Florida

B.S. Computer Engineering, Dean's List, Gold's Scholars | GPA: 3.52/4.00

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OPEN SOURCE

Open Source Test and Reliability Fixes | C++, Go, Python, C, GoogleTest/gmock, pytest, Docker, Gerrit and GitHub code review

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- **OpenBMC phosphor-user-manager:** root-caused a reported 60-of-69 test failure to a missing test-environment dependency rather than a code defect, then found and fixed two real bugs uncovered once the environment was corrected, an unhandled exception that could abort setup for every user over one bad record, and a C++ construction-order bug that let unit tests silently run against the real system instead of their mocks; verified with full before-and-after test counts across both root and non-root runs.
- **Zephyr RTOS:** found a devicetree macro bug in the MIPI DBI display driver, proved it with a targeted compile-time check against an existing in-tree test fixture, fixed it, and added a permanent regression test so it cannot silently reappear.
- **TinyGo:** identified an unsynchronized write racing against every other lock-protected access to the same runtime data, built a targeted concurrency stress test to try to reproduce the reported crash, and reported the negative repro result honestly alongside the fix rather than overstating what the test showed.
- **PlatformIO:** added a regression test reproducing a device-monitor crash before fixing it, then confirmed the project's full lint suite (black, isort, codespell, pylint) passed clean; merged upstream.

PROJECTS

Testbench | Python, MQTT, Docker, GitHub Actions, MSP430FR2355, ESP32, STM32F103C8, RP2040

Jul

- Built a hardware-agnostic test automation platform: a capture-and-replay tool and hardware simulator let the full test suite run without physical devices attached.
- Wrote reference firmware for four different microcontrollers emitting a shared JSON-over-UART frame format, plus a schema-validated parser and a cross-platform test suite asserting every board's output against that schema.
- Set up a GitHub Actions workflow that spins up Postgres and Mosquitto containers and runs an end-to-end simulated-hardware regression test on every push.

Radiology Second-Opinion Agent | FastAPI, Docker Compose, MLflow, Evidently, pytest

Jul

- Data and MLOps engineer on a 4-person hackathon team; owned the model serving layer and its test coverage, including unit tests, mock-LLM tests, and endpoint validation, plus Evidently-based data drift monitoring in production.

FPGA Audio Message Recorder | Xilinx Spartan-6, Verilog, PicoBlaze, DDR2 SDRAM, I2C, UART

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- Collaborated with a 3-member team to design, synthesize, and deploy an audio recording and playback system on a Xilinx Spartan-6 FPGA, backed by a DDR2 SDRAM interface and a PicoBlaze soft-core for the UART menu.
- Debugged and fixed real hardware issues, including audio distortion and a rewind pointer bug, verifying each fix against the physical board across several controller revisions.

EXPERIENCE

City of Winter Haven Technology Services

Smart City Student Volunteer

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- Installed server hardware and performed rack maintenance for city technology infrastructure.
- Provisioned and configured YubiKey security keys for staff authentication.
- Resolved IT support tickets and optimized PyTorch/TensorFlow environments for the city's early-stage AI hosting infrastructure.

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Manager, Shift Leader, and Front-End Web Support

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- Developed the front-end of the restaurant's Square-based ordering site, working within the platform's UI constraints to deliver a functional ordering experience.
- Managed shift operations and enforced food safety procedures.

INVOLVEMENT

Member, Society of Hispanic Professional Engineers (SHPE), Society of Asian Scientists and Engineers (SASE), and IEEE